



ICRS Robotics 101

Session 1 (Mechanical)

So, you want to build a thing?

- Search other designs (Reddit, Google, Discord)
- What you want has a name, make sure you figure it out
- Even better is talk to someone that's done it before

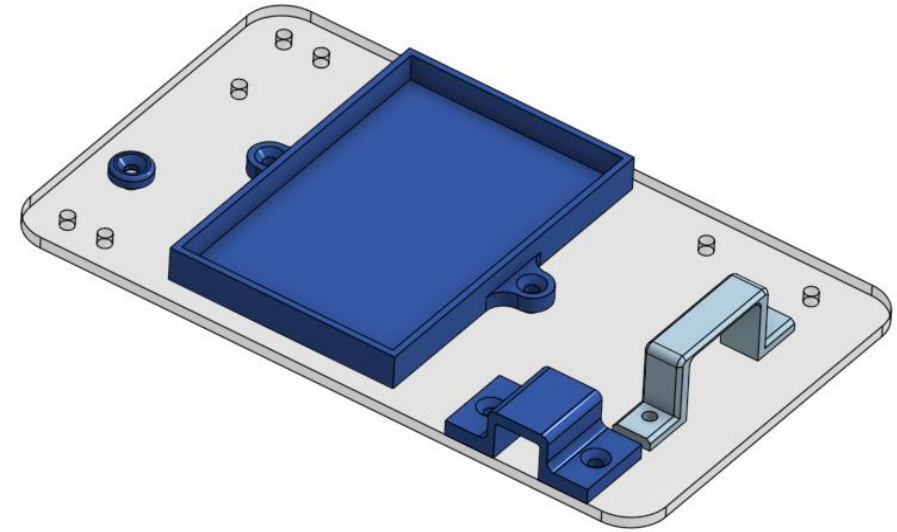


Mechanical Design

- You don't need to know material properties are tensors to build something that won't break
- "Good practices" exist and exist for a reason
- You get better by doing

General mechanical tips

- Break design into multiple parts
- Use off the shelf stuff whenever possible
- Stiffness is usually more important than strength
- Think about assembly

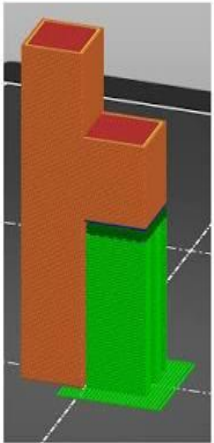


Design for Manufacturing (DFM)

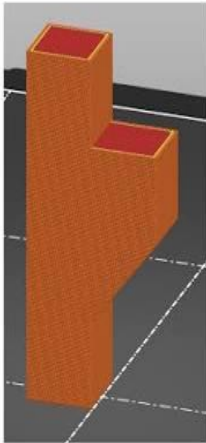
- Start with 1 method and material:
 - Metal machining
 - 3d printing
 - laser cutting
- Get better at it:
 - milling -> CNC
 - advanced slicer settings/materials
 - Thermoforming (bending like sheet metal)
- Again, talk to people that have done it before as they will give you the best insight

3D printing DFM

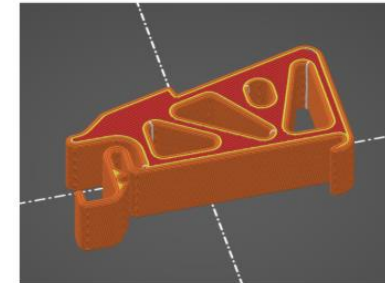
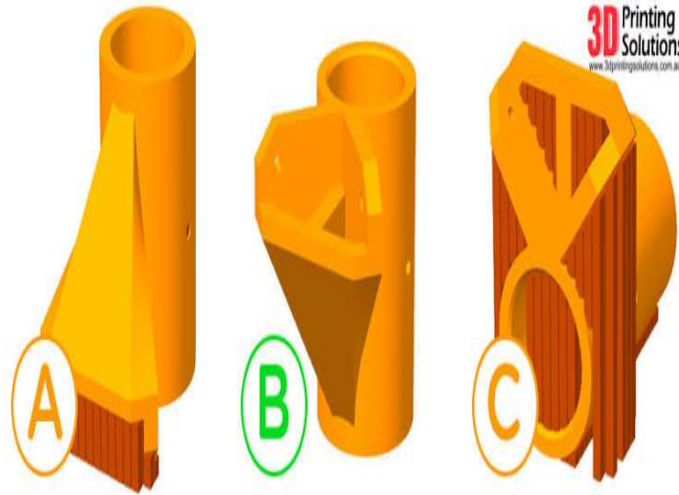
- Official term is fused deposition modelling (FDM)



not ideal

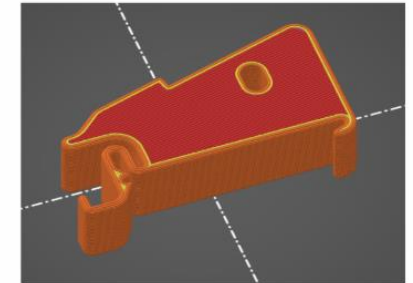


improved



Sliced Info	
Used Filament (g)	4.63 (197.63)
(including spool)	
Used Filament (m)	1.52
Used Filament (mm³)	3645.50
Cost	0.17
Estimated printing time:	
- normal mode	20m

4.63g 20min



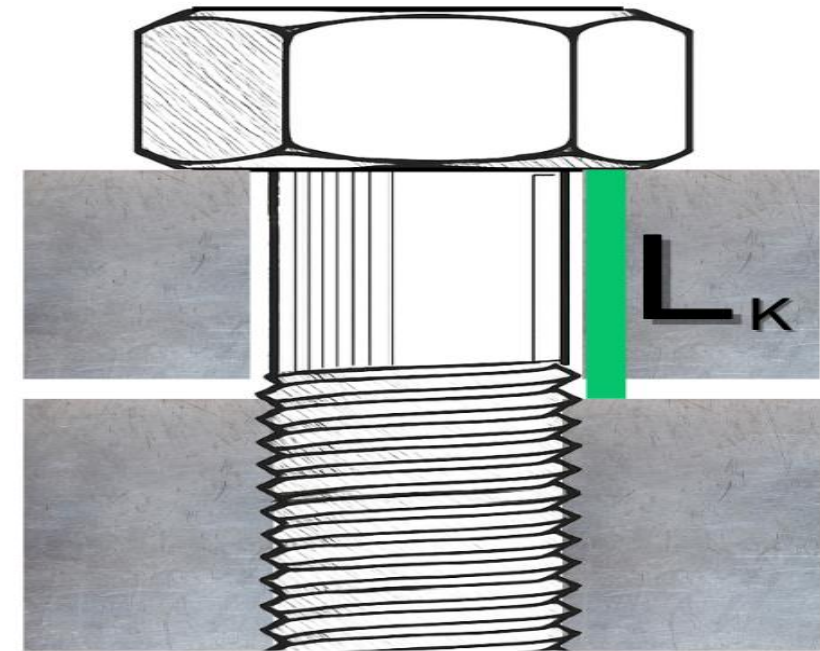
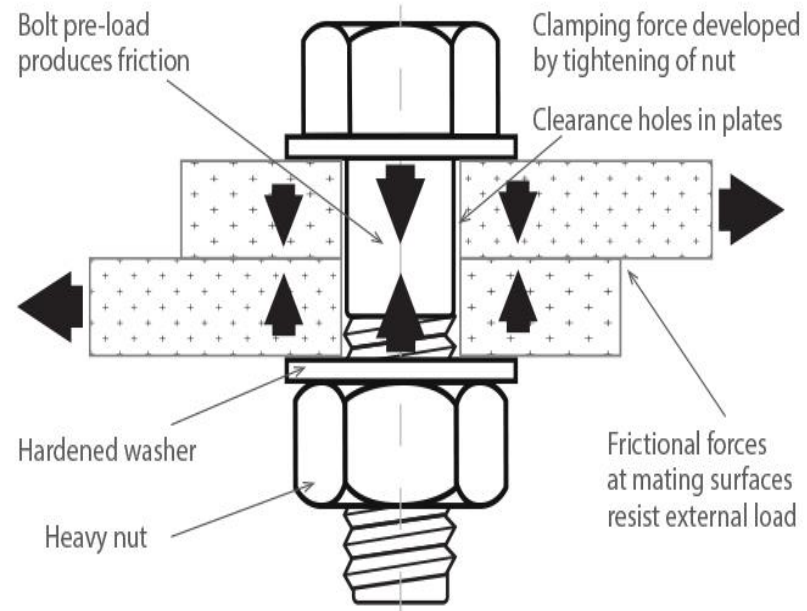
Sliced Info	
Used Filament (g)	4.35 (197.35)
(including spool)	
Used Filament (m)	1.42
Used Filament (mm³)	3423.04
Cost	0.16
Estimated printing time:	
- normal mode	15m

4.35g 15min

Tolerances and fits

- 3 kinds of fits: Clearance, Interference Transition
- An M3 bolt wants a 3.2mm hole
- 3D printing is fast but not accurate

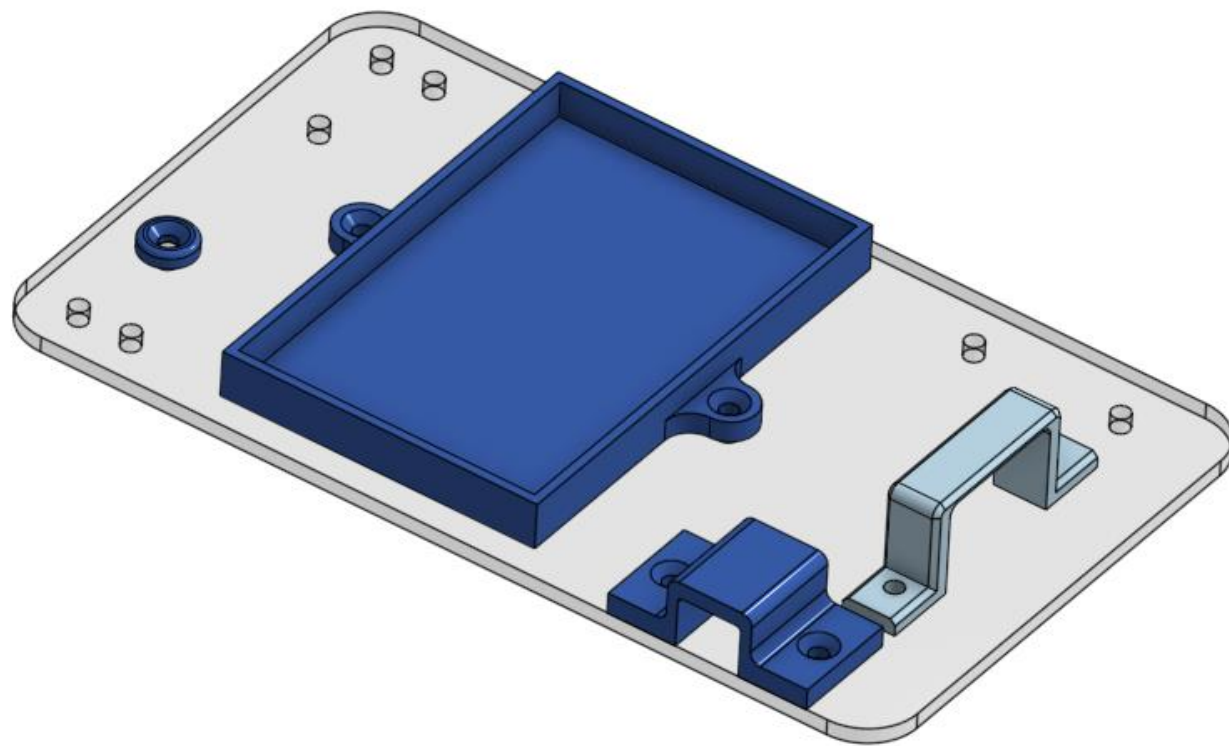
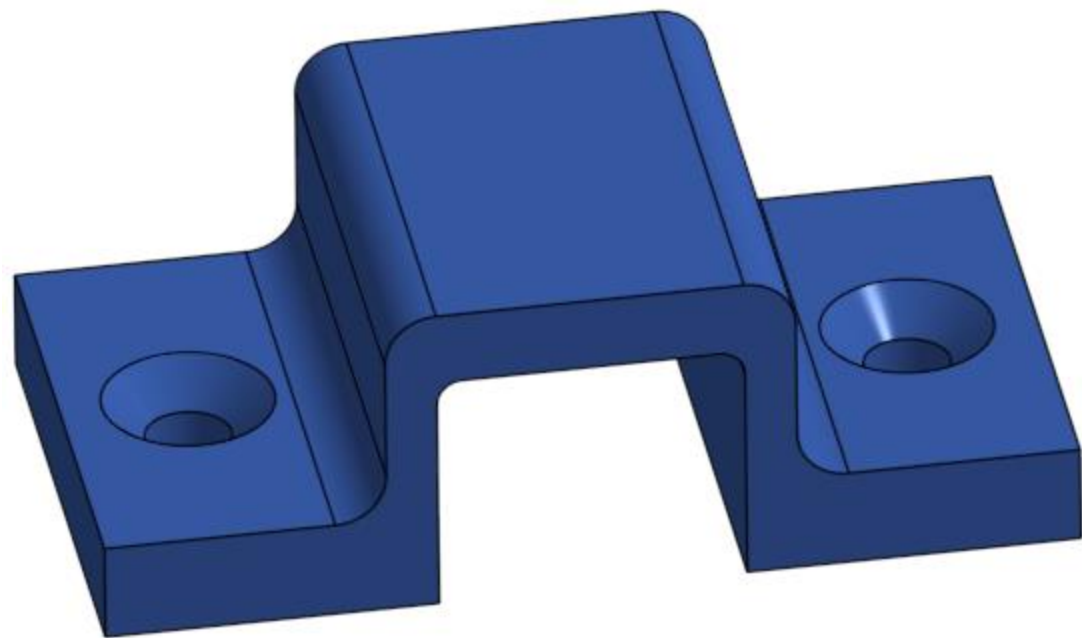
Fasteners

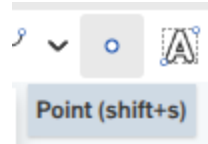
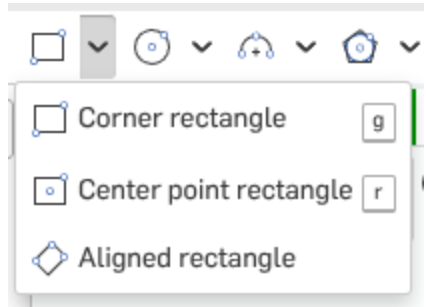


Bolt in blind hole

Onshape

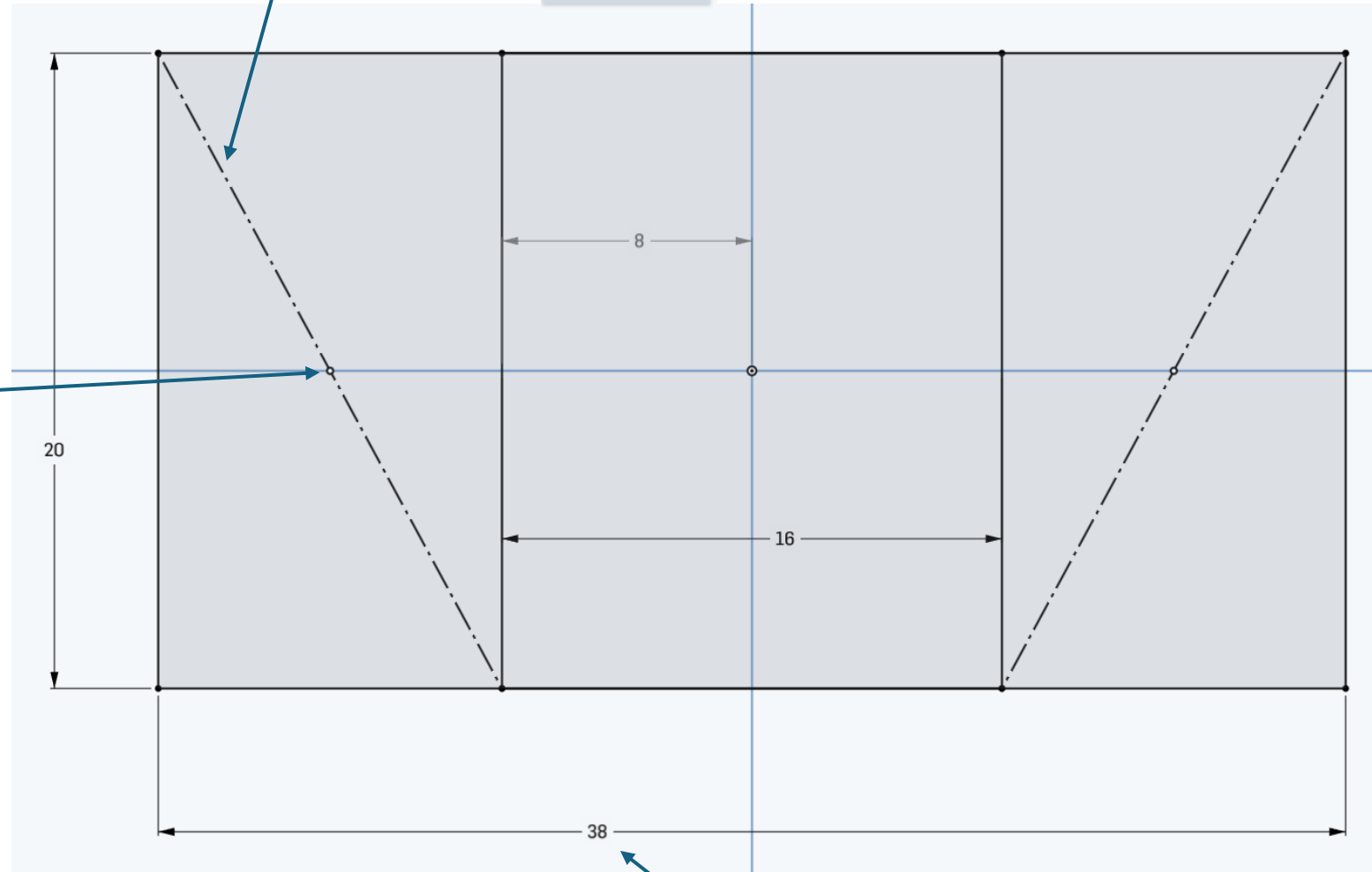
- Link: <https://www.onshape.com/en/sign-up>





Point

Construction line



Dimension



tch



Extrude (shift+e)

Extrude 1  

Solid

Surface

Thin

New

Add

Remove

Intersect

Faces and sketch regions to extrude

Face of Sketch 1

×

Face of Sketch 1

×

Blind  

> Depth

4 mm



> ☐ Direction

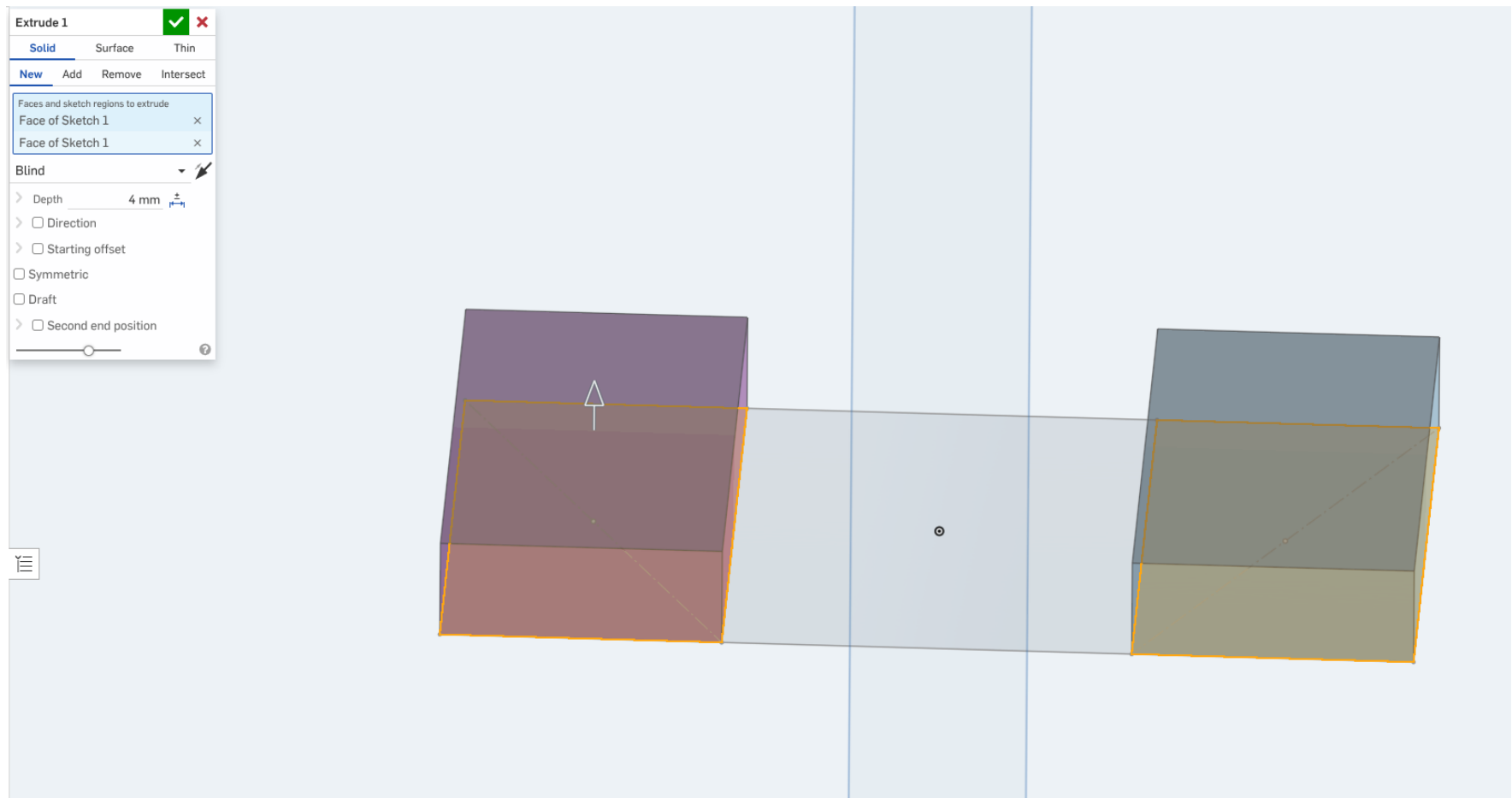
> ☐ Starting offset

☐ Symmetric

☐ Draft

> ☐ Second end position





Extrude 2

✓

✗

Solid

Surface

Thin

New

Add

Remove

Intersect

Faces and sketch regions to extrude

Face of Sketch 1

✕

Blind

▼

✎

> Depth

13 mm

±

↔

> ☐ Direction

> ☐ Starting offset

☐ Symmetric

☐ Draft

> ☐ Second end position

☐ Merge with all

Merge scope

Part 2

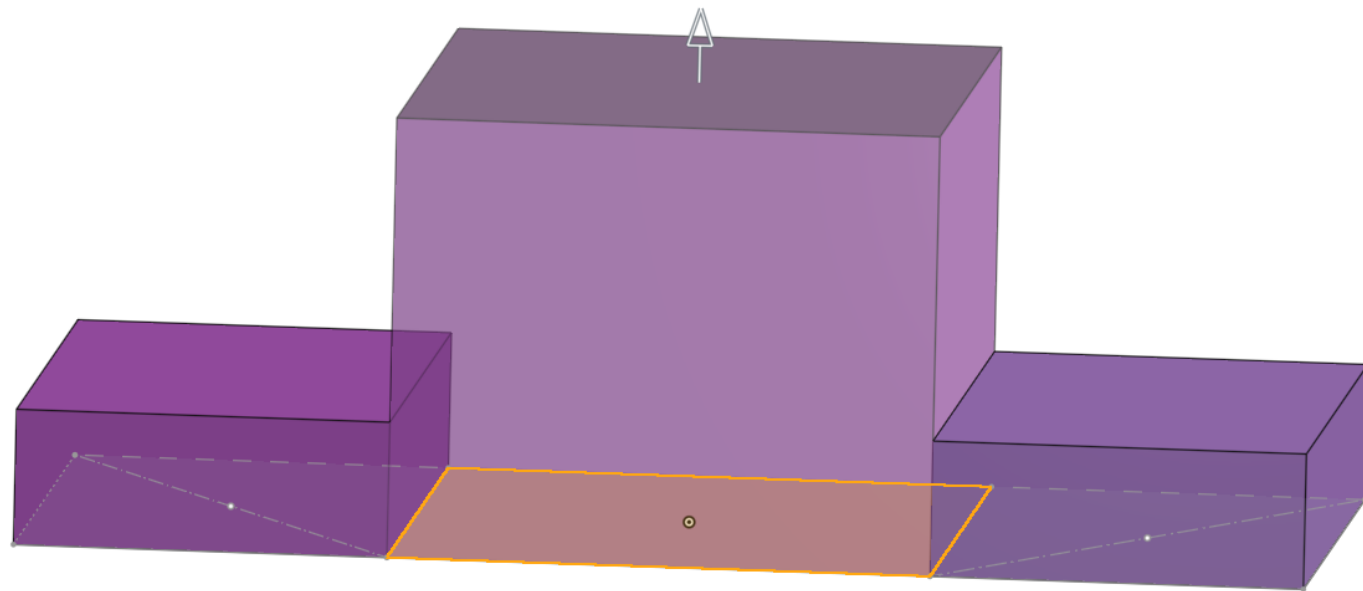
✕

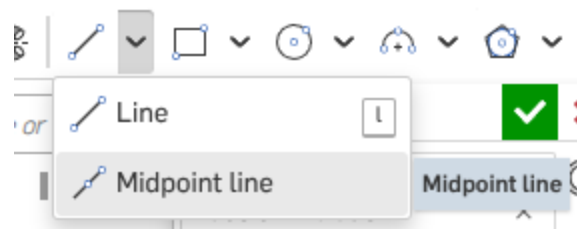
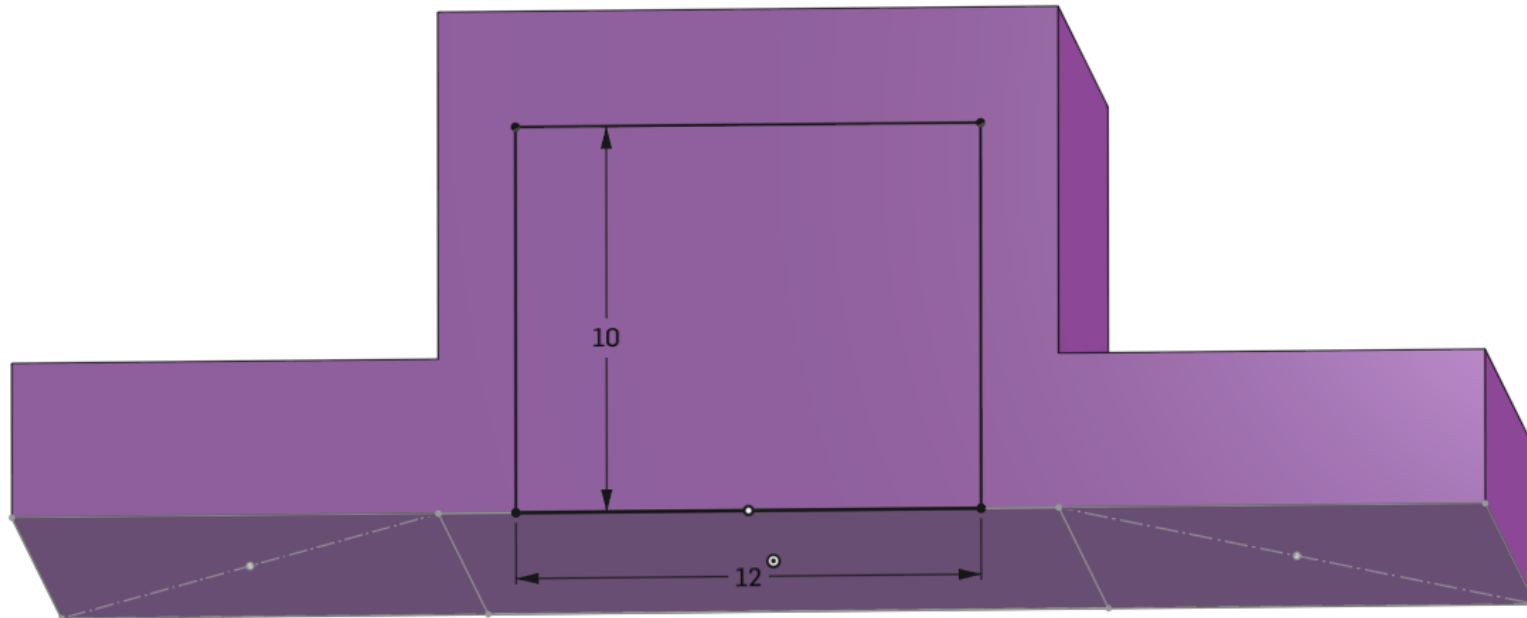
Part 1

✕

—○—

?





Extrude 3

✓

✕

Solid

Surface

Thin

New

Add

Remove

Intersect

Faces and sketch regions to extrude

Face of Sketch 2

✕

Up to face

▼

↗

Up to face

Face of Extrude 1

✕

🕒

☐ Offset distance

> ☐ Direction

> ☐ Starting offset

☐ Draft

> ☐ Second end position

☐ Merge with all

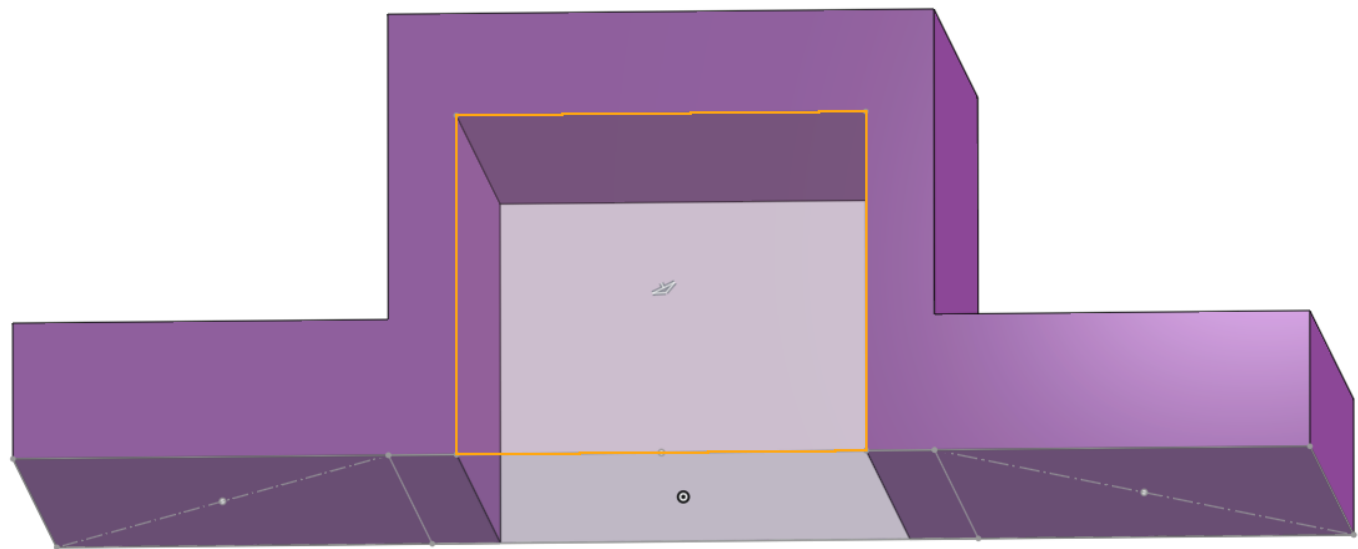
Merge scope

Part 1

✕

?

☰



Fillet 1 ✓ ✗

Edge Full round

Entities to fillet

Edge of Extrude 2	✕
Edge of Extrude 2	✕
Edge of Extrude 2	✕
Edge of Extrude 2	✕

☒ Tangent propagation

Measurement **Radius**

Control **Distance**

> Radius **2 mm**

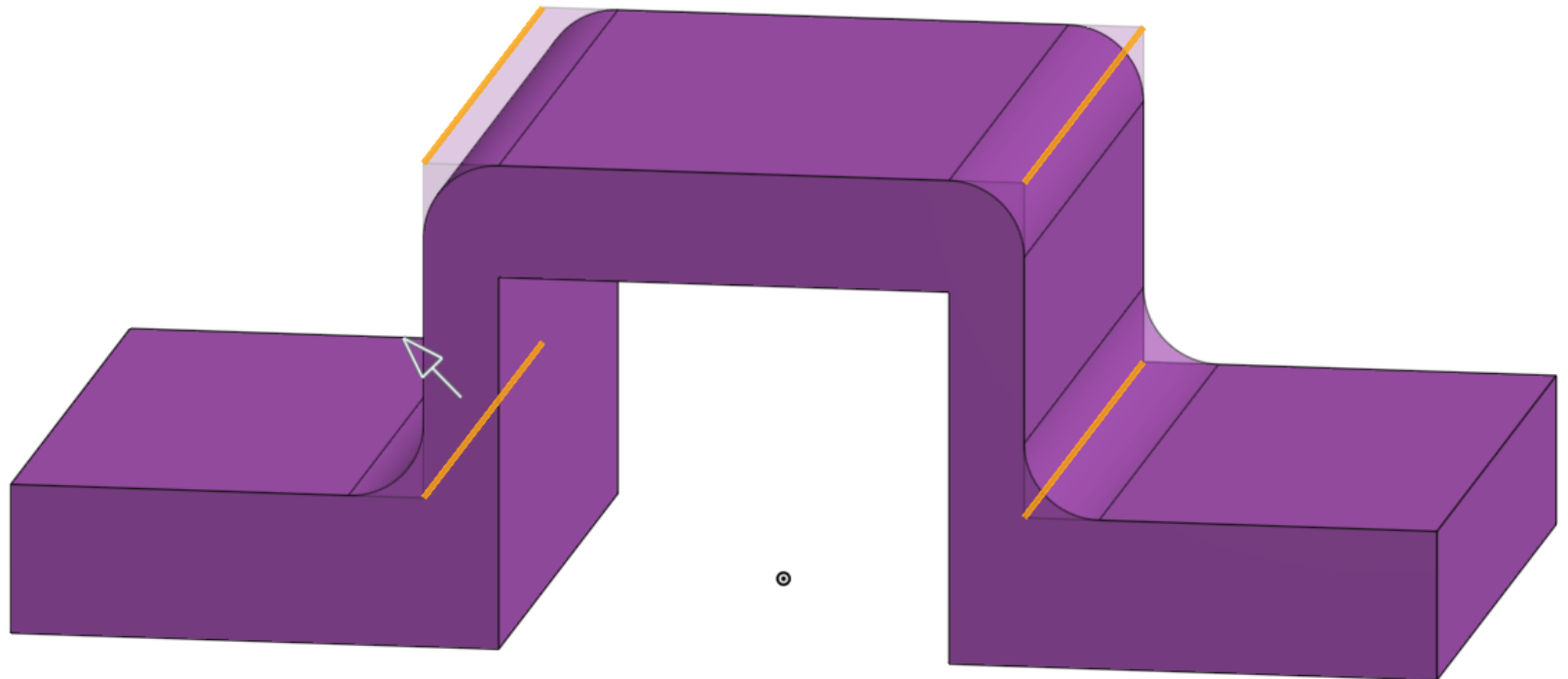
☐ Asymmetric

> ☐ Partial fillet

> ☐ Variable fillet

> ☒ Allow edge overflow

> ☐ Smooth fillet corners



Fillet 2 ✓ ✗

Edge Full round

Entities to fillet	
Edge of Extrude 3	✕
Edge of Extrude 3	✕

☒ Tangent propagation

Measurement Radius ▼

Control Distance ▼

> Radius 1 mm $\frac{\pm}{\mu\text{m}}$

☐ Asymmetric

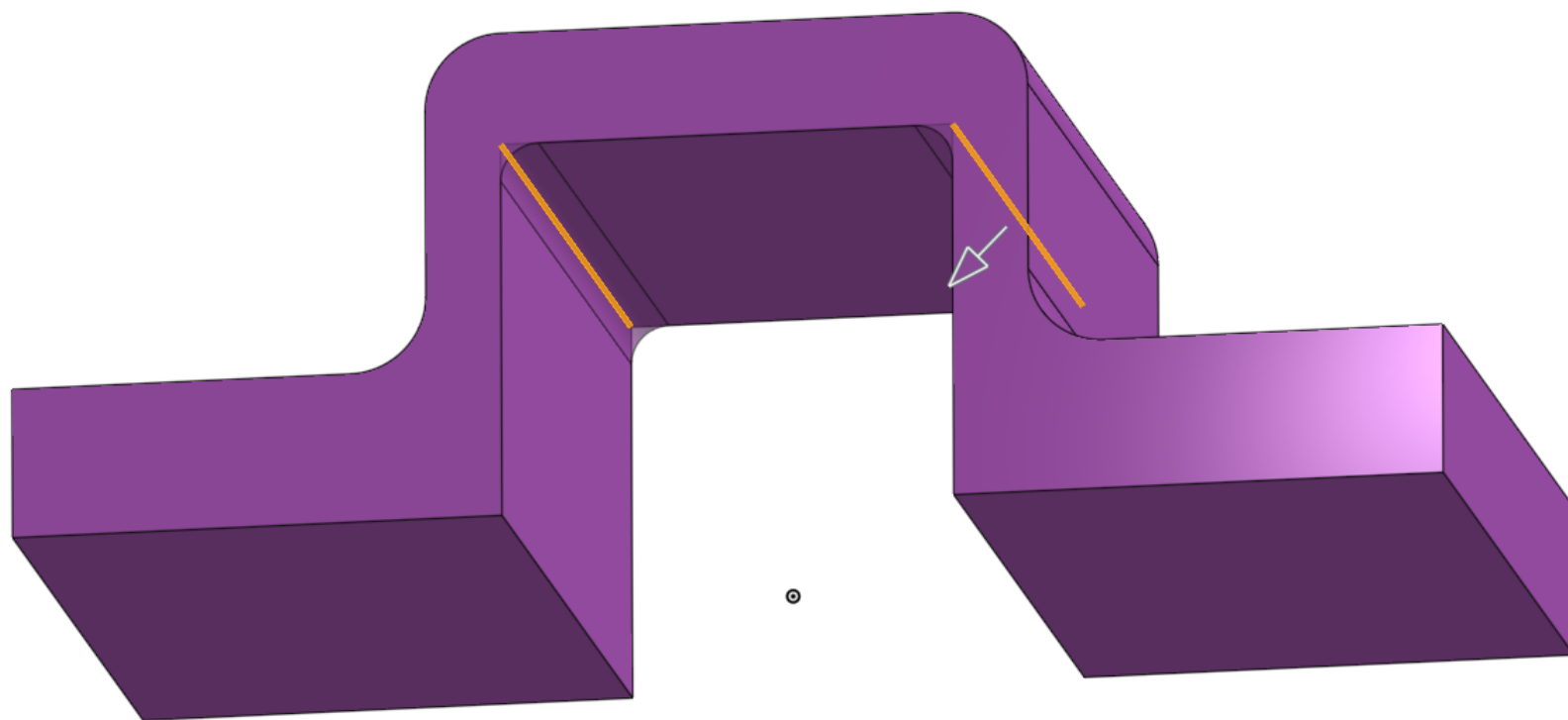
> ☐ Partial fillet

> ☐ Variable fillet

> ☒ Allow edge overflow

> ☐ Smooth fillet corners

— — ?



M3x0.50 ⚡ 12 mm ✓ ✗

Inch

Metric

Simple

Counterbore

Countersink

Sketch points to place holes

Vertex of Sketch 1 ✕

Vertex of Sketch 1 ✕

Merge scope

Part 1 ✕

Hole type

Tapped

Tap type

Straight tap

Size

M3

Pitch

0.50 mm (Coar...

Fastener fit

None

>

☐ Thread class

>

 2.5 mm

Start plane

Start from selected plane

Hole start plane or mate conne...

Face of Extrude 1 ✕

Termination

Blind

>

 12 mm

>

 118 deg

>

 12 mm

>

 0



?

